

15. Uncertainty budget, limitations, and open questions

The smallest Z steps and the absolute bias are the measurements most limited by the setup. The relative measurements are robust: noise, rise distance, transfer-curve shape, and detection minimums. The magnitudes below are typical values, not measured ones; replace them with measured values after Step 4.

Source	Estimated magnitude	Affects	Mitigation
Robot repeatability	0.02–0.05 mm	Smallest B-Z steps; edge position	Approach every Z visit from the same direction; A, B, A, B averaging over the cycles; the ratio of the repeatability to each rung reported with δ_{50} as the truth uncertainty; averaging over phase poses
Robot absolute accuracy	0.2–1 mm over large moves	bias(Z), registration	Plane-correspondence registration over REGISTRATION_POSES poses spanning the full Z range; tests run as local moves from registered stations
Registration residual	≤ 0.15 mm RMS (acceptance limit)	bias(Z), edge offset s_{50} , small-feature area	Acceptance gate (Step 4.7); report the residual with every result
Plane-registration depth offset	about 0.1 mm (rough estimate: the plane-fit error amplified by $1/(1 - \cos 20^\circ) \approx 17$ at the tilt limit, averaged over the poses)	Absolute bias(Z) only	Many poses over the full Z range; tilts to the limit of REGISTRATION_TILT_RANGE_DEG; residual reported with every result
As-built diameter uncertainty	set by the measuring instrument	A_{true} : relative error $\approx 2\delta D/D$, for example 0.6% at 7 mm with $\delta D = 0.02$ mm	Optical comparator or microscope; carry δD into the area and D_0 confidence intervals
Plate flatness	≤ 0.05 mm	Fixed-pattern noise, bias	Flatness spec; map the plate on a CMM if available
Thermal drift	unknown until the sentinels run	bias(Z), B-Z	Warm-up gate, sentinels, ABAB ordering, randomized station order
Front-back interreflection	unknown	Cutout and small-gap results	Compare G small vs large; matte finish
Sensor temporal filter	depends on configuration	σ_t (underestimated), trial independence	

			Discard frames after each move until the plane mean settles (set in Step 4.4); run filters off if possible
Reflectance mismatch	avoided by design	Edges, area	One finish on all surfaces

Limitations. Results hold for one surface finish, static targets, mostly fronto-parallel poses, and the sensor configuration recorded in `sensor_config.json`. Registration from planes leaves the camera's depth offset conditioned by the registration tilt range (table above), so absolute bias(Z) carries that uncertainty; Z -scale and nonlinear bias do not, because a rigid transform cannot absorb them. The 0% point is an extrapolation of the fitted detection curve below the lowest measured point, D_5 , and is reported as a prediction, not a measurement. Because the read-back robot pose is the step truth of B- Z , the smallest detectable Z step for the larger patches may come out as a bound set by the robot's pose-reporting accuracy rather than a measurement. The range limits 400 and 1600 mm assume the sensor reads there; Step 4.5 confirms this before any series runs.

Open questions

- ☐ VSX3000 f_x , f_y , baseline, projector offset, FOV, depth LSB, and frame rate in the chosen trigger mode. These fill the $\dagger$ parameters, the chamfer check, and the time budget.
- ☐ Which on-sensor processing configuration is the production one?
- ☐ Which surface finish represents the production targets, and is a second finish needed?